Customer Churn Analysis & Retention Dashboard

Data Science & Analytics – Task 2

Prepared By:
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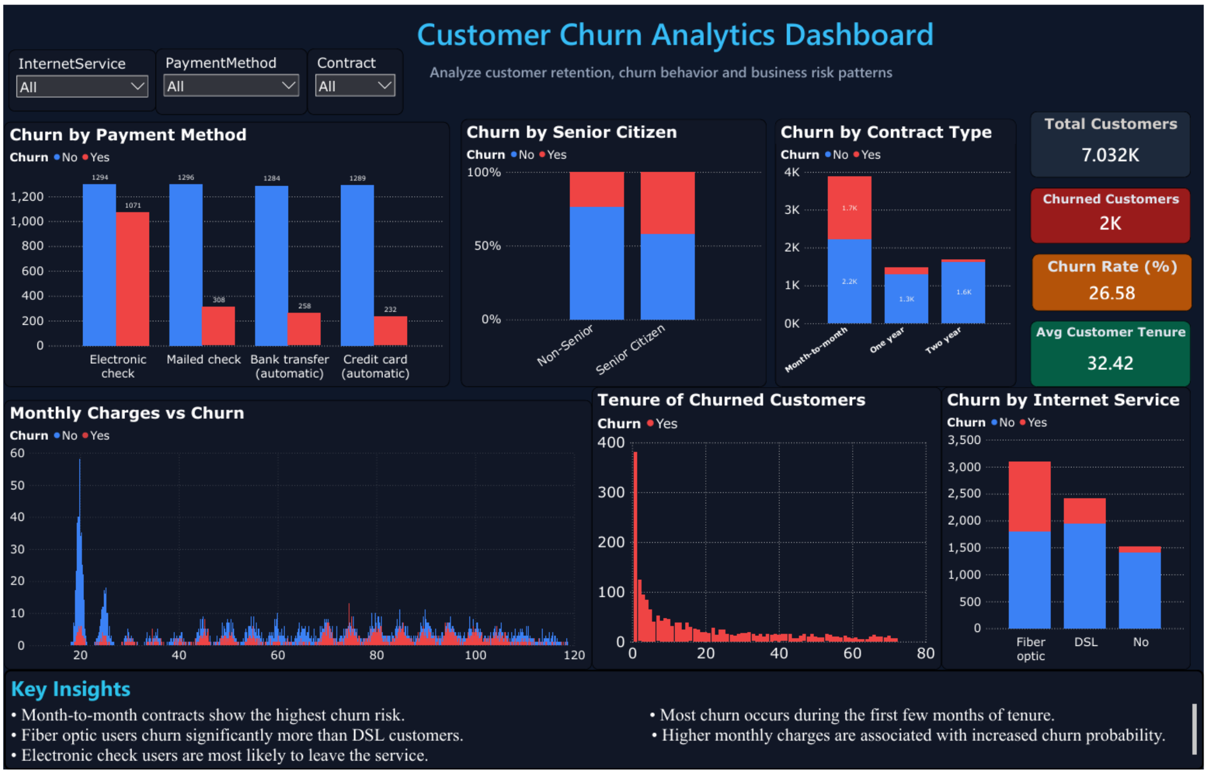
Submitted To:
Future Interns

Tools Used:
Python (Google Colab), Power BI, CSV Dataset

Dataset Used:
Telco Customer Churn Dataset



Project Overview

Customer churn is one of the most significant challenges faced by subscription-based businesses, SaaS companies, telecommunication providers, and digital platforms. Customer churn refers to the percentage of customers who stop using a service over a given period.

Understanding customer behavior and identifying churn factors is essential because retaining existing customers is often more cost-effective than acquiring new ones. Businesses can improve customer satisfaction, reduce revenue loss, and optimize services through churn analysis.

This project focuses on analyzing customer churn behavior using the Telco Customer Churn Dataset. The analysis was conducted using Python in Google Colab for exploratory data analysis and Power BI for dashboard visualization.

The objective of this project was to identify customer segments with high churn probability, analyze retention patterns, understand factors affecting churn, and provide actionable business recommendations to improve customer retention.

The final outcome of this project is an interactive business dashboard that enables stakeholders to understand churn trends and make data-driven decisions.

Project Objectives

The main objectives of this project are:

1. Analyze Customer Churn

To identify the percentage of customers leaving the service and understand overall churn behavior.

2. Understand Retention Patterns

To study how long customers stay with the company and identify customer lifetime trends.

3. Identify Churn Drivers

To analyze how contract type, internet service, payment methods, monthly charges, and senior citizen status influence churn.

4. Customer Segmentation

To identify which customer groups are more likely to churn.

5. Build Interactive Dashboard

To create a professional dashboard that presents business insights clearly for stakeholders.

6. Generate Business Recommendations

To suggest actionable strategies that can reduce churn and improve customer retention.

Dataset Information

Dataset Name:

Telco Customer Churn Dataset

Dataset Source:

Kaggle

Total Records:

7,043 Customers

Total Features:

21 Columns

Dataset Description

The dataset contains customer-related information from a telecommunications company. It includes demographic information, subscription details, payment methods, contract information, monthly billing details, and churn status.

The target variable in this dataset is **“Churn”**, which indicates whether a customer left the service or stayed.

Important Features in Dataset

- Customer ID
- Gender
- Senior Citizen Status
- Partner Status
- Dependents
- Tenure
- Internet Service
- Contract Type

- Payment Method
- Monthly Charges
- Total Charges
- Churn Status

Data Quality

The dataset was examined for missing values, duplicate records, and data inconsistencies.

Observations:

- Total rows: 7,043
- Total columns: 21
- Missing values: None
- Dataset quality: Clean and suitable for analysis

The dataset required minimal preprocessing before visualization and analysis.

Tools & Technologies Used

The following tools and technologies were used to complete this project:

Tool/Technology	Purpose
Python	Data analysis and preprocessing
Google Colab	Python execution environment
Pandas	Data cleaning and manipulation
Matplotlib	Data visualization
Power BI	Interactive dashboard development
CSV Dataset	Data source for analysis

Why These Tools Were Used

Python & Pandas were used for performing data preprocessing and exploratory data analysis.

Matplotlib was used to create charts for churn distribution and behavioral analysis.

Power BI was used to build a visually interactive business dashboard for stakeholders.

The combination of Python and Power BI helped integrate both technical analysis and business storytelling.

Data Cleaning & Preprocessing

Before performing analysis, the dataset was cleaned and examined to ensure accuracy and consistency.

The following preprocessing steps were performed:

Step 1: Dataset Import

The CSV dataset was imported into Google Colab using Pandas.

Step 2: Dataset Shape Analysis

The number of rows and columns in the dataset was examined.

Result:

- Rows: 7,043
- Columns: 21

Step 3: Missing Value Analysis

The dataset was checked for null or missing values.

Result:

- No missing values were found.

Step 4: Data Type Inspection

Data types of all columns were analyzed to understand categorical and numerical variables.

Step 5: Churn Distribution Analysis

The target variable "Churn" was analyzed.

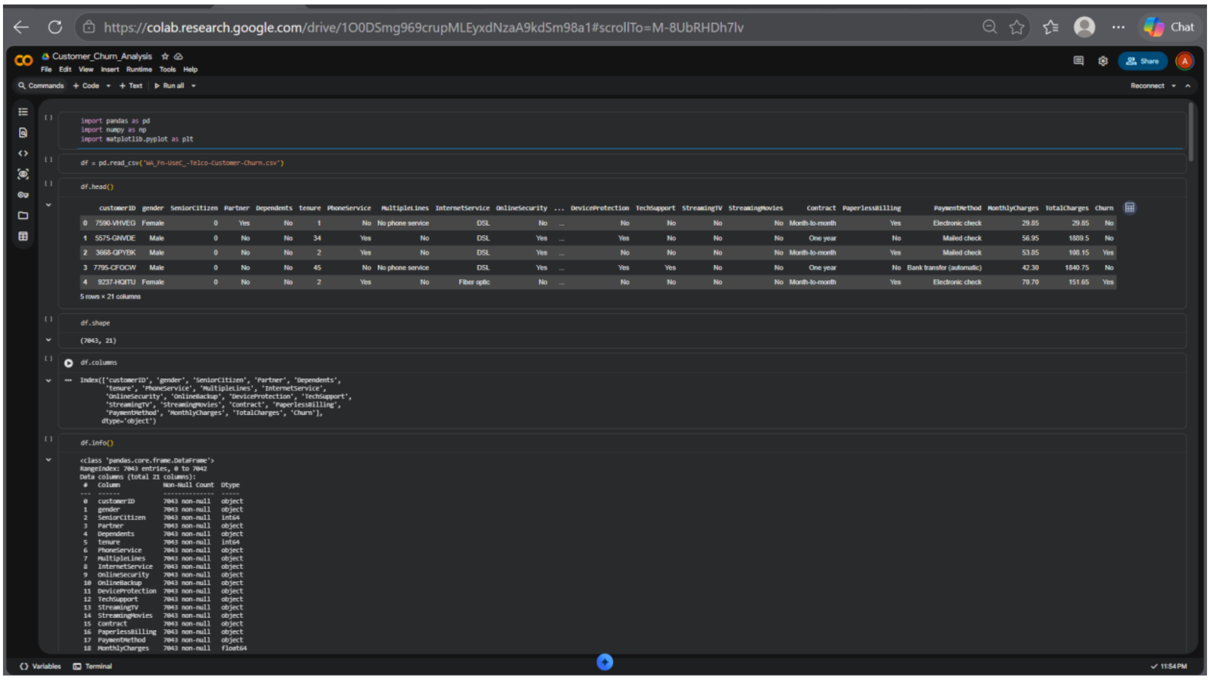
Results:

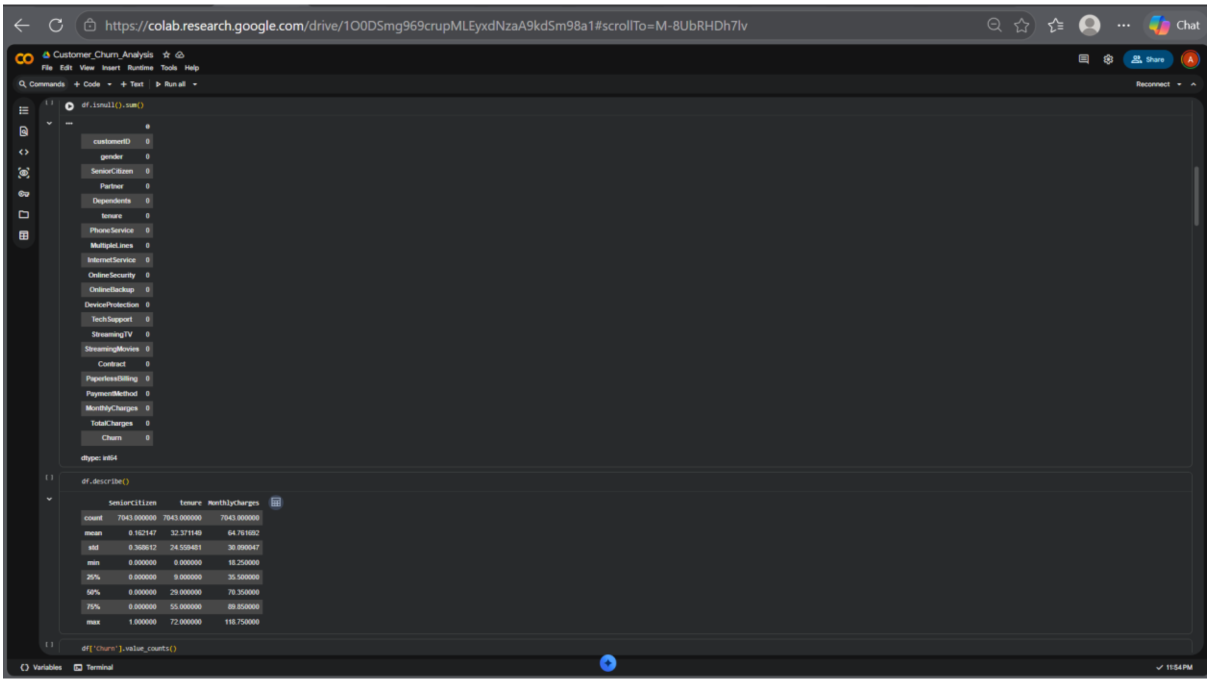
- Customers Stayed: 5,174 (73.46%)
- Customers Churned: 1,869 (26.54%)

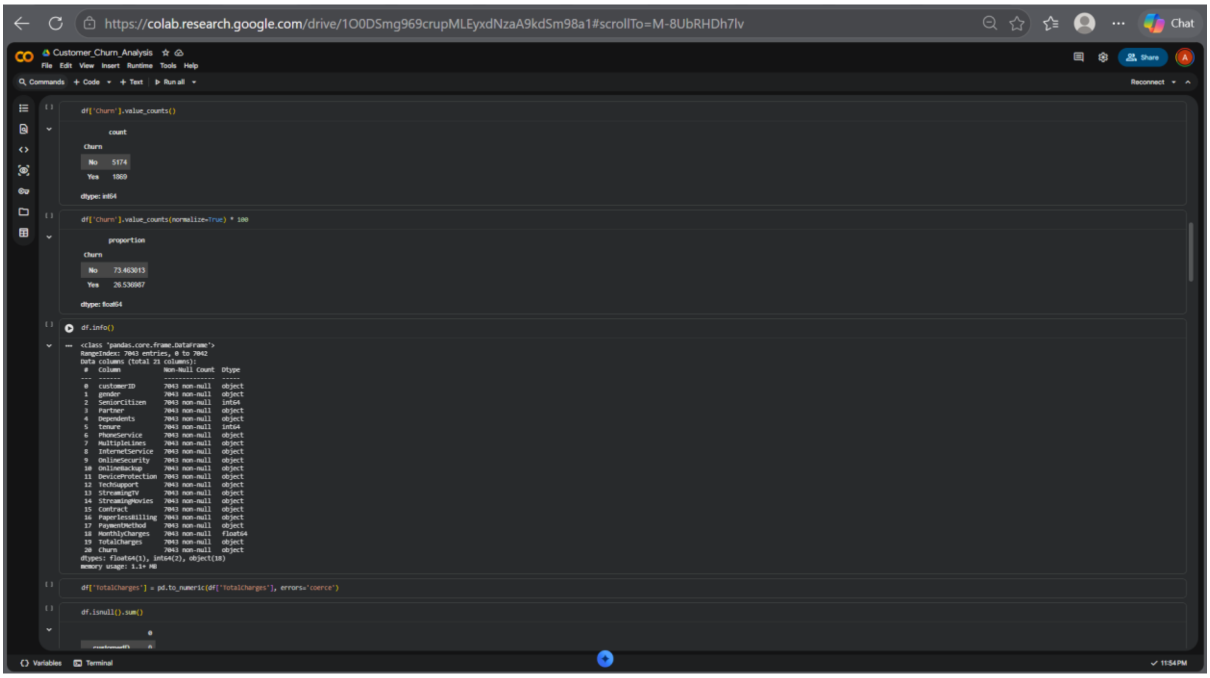
This indicates that nearly one-fourth of the customers discontinued the service.

Step 6: Data Preparation for Dashboard

The cleaned dataset was exported and imported into Power BI for dashboard creation.







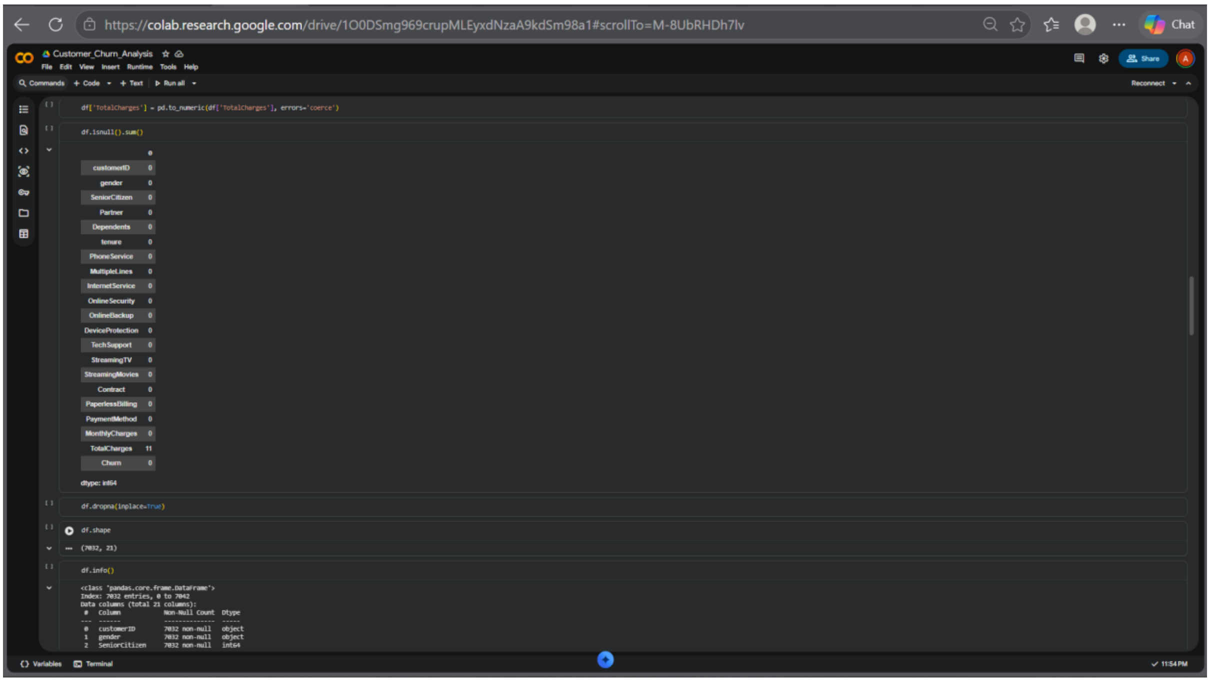
```
df['Churn'].value_counts()

      count
Churn
No     5174
Yes    1869

dtype: int64
```

```
df['Churn'].value_counts(normalize=True) * 100

      proportion
Churn
No     73.463013
Yes    26.536987

dtype: float64
```

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
```

```
df.isnull().sum()
```



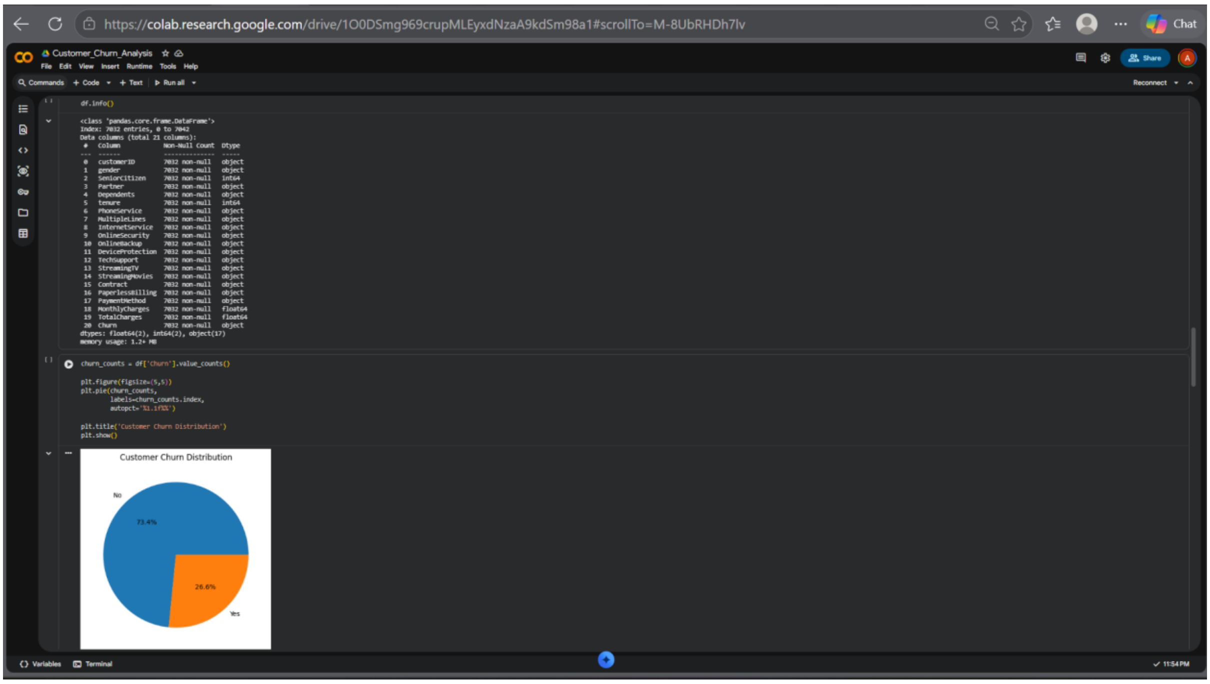
```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
```

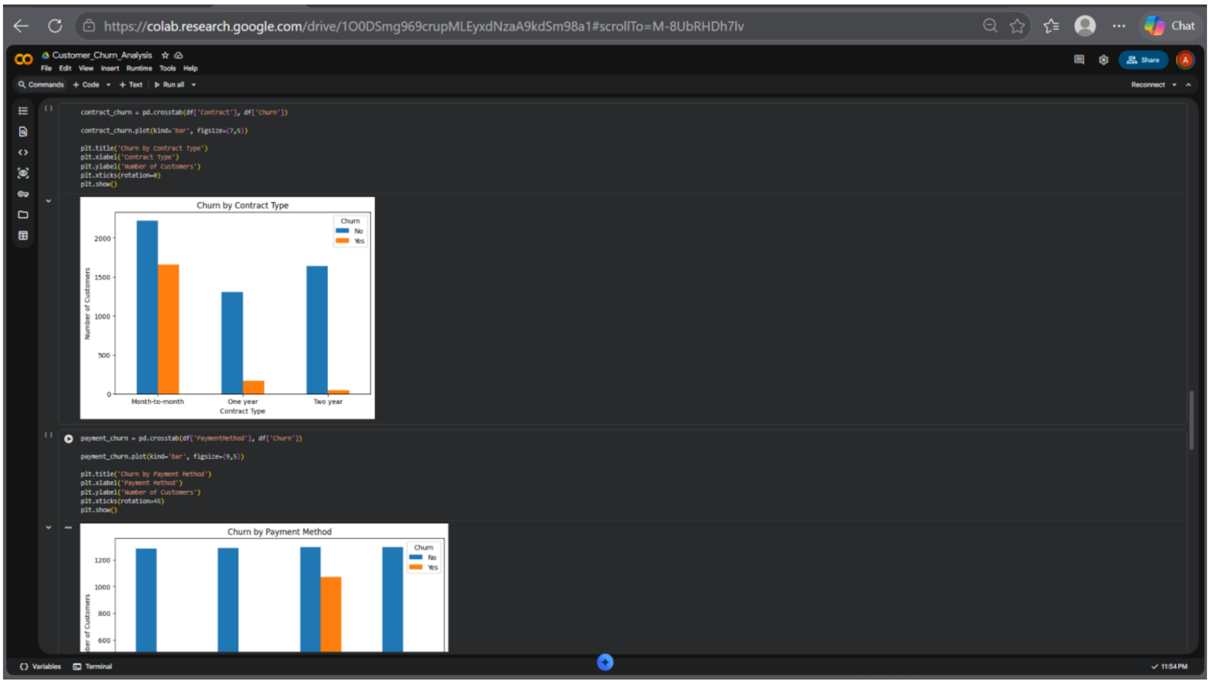
```
df.isnull().sum()

                  0
customerID        0
gender            0
SeniorCitizen     0
Partner           0
Dependents        0
tenure            0
PhoneService      0
MultipleLines     0
InternetService   0
OnlineSecurity    0
OnlineBackup      0
DeviceProtection  0
TechSupport       0
StreamingTV       0
StreamingMovies   0
Contract          0
PaperlessBilling  0
PaymentMethod     0
MonthlyCharges    0
TotalCharges     11
Churn             0

dtype: int64
```

```
df.dropna(inplace=True)
```

```
df.shape

(7032, 21)
```

```
df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 7032 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7032 non-null   object
 1   gender            7032 non-null   object
 2   SeniorCitizen     7032 non-null   int64
```


https://colab.research.google.com/drive/1O0DSmg969crupMLEyxdNzaA9kdSm98a1#scrollTo=M-8UbRHDh7lv
Customer_Churn_Analysis
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
df.info()
<class 'pandas.core.frame.DataFrame'>
Index: 7032 entries, 0 to 7042
Data columns (total 21 columns):
Column Non-Null Count Dtype
0 customerID 7032 non-null object
1 gender 7032 non-null object
2 SeniorCitizen 7032 non-null int64
3 Partner 7032 non-null object
4 Dependents 7032 non-null object
5 tenure 7032 non-null int64
6 PhoneService 7032 non-null object
7 MultipleLines 7032 non-null object
8 InternetService 7032 non-null object
9 OnlineSecurity 7032 non-null object
10 OnlineBackup 7032 non-null object
11 DeviceProtection 7032 non-null object
12 TechSupport 7032 non-null object
13 StreamingTV 7032 non-null object
14 StreamingMovies 7032 non-null object
15 Contract 7032 non-null object
16 PaperlessBilling 7032 non-null object
17 PaymentMethod 7032 non-null object
18 MonthlyCharges 7032 non-null float64
19 TotalCharges 7032 non-null float64
20 Churn 7032 non-null object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.2+ MB
churn_counts = df['Churn'].value_counts()
plt.figure(figsize=(5,5))
plt.pie(churn_counts,
labels=churn_counts.index,
autopct='%1.1f%%')
plt.title('Customer Churn Distribution')
plt.show()
Customer Churn Distribution
No
73.4%
26.6%
Yes
Variables
Terminal
11:54 PM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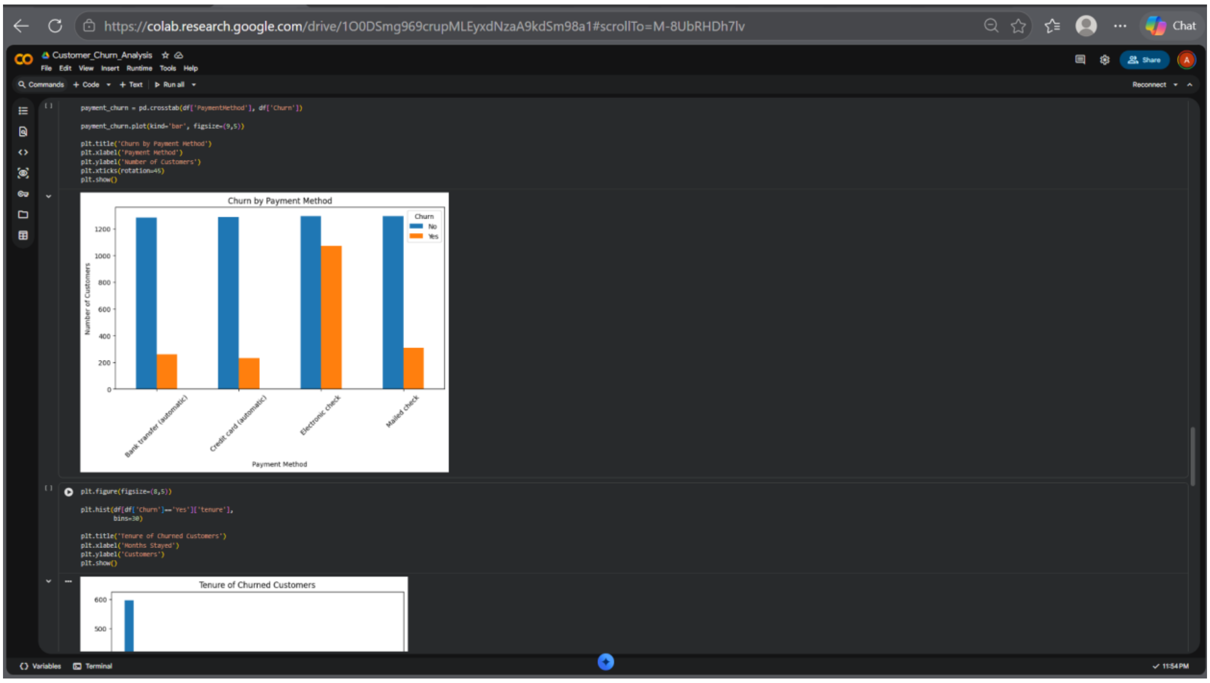
Customer_Churn_Analysis
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
contract_churn = pd.crosstab(df['Contract'], df['Churn'])
contract_churn.plot(kind='bar', figsize=(7,5))
plt.title('Churn by Contract Type')
plt.xlabel('Contract Type')
plt.ylabel('Number of Customers')
plt.xticks(rotation=0)
plt.show()
Churn by Contract Type
Churn
No
Yes
Number of Customers
2000
1500
1000
500
0
Month-to-month
One year
Two year
Contract Type
payment_churn = pd.crosstab(df['PaymentMethod'], df['Churn'])
payment_churn.plot(kind='bar', figsize=(9,5))
plt.title('Churn by Payment Method')
plt.xlabel('Payment Method')
plt.ylabel('Number of Customers')
plt.xticks(rotation=45)
plt.show()
Churn by Payment Method
Churn
No
Yes
1200
1000
800
600
Variables
Terminal
11:54 PM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Chat
Customer_Churn_Analysis
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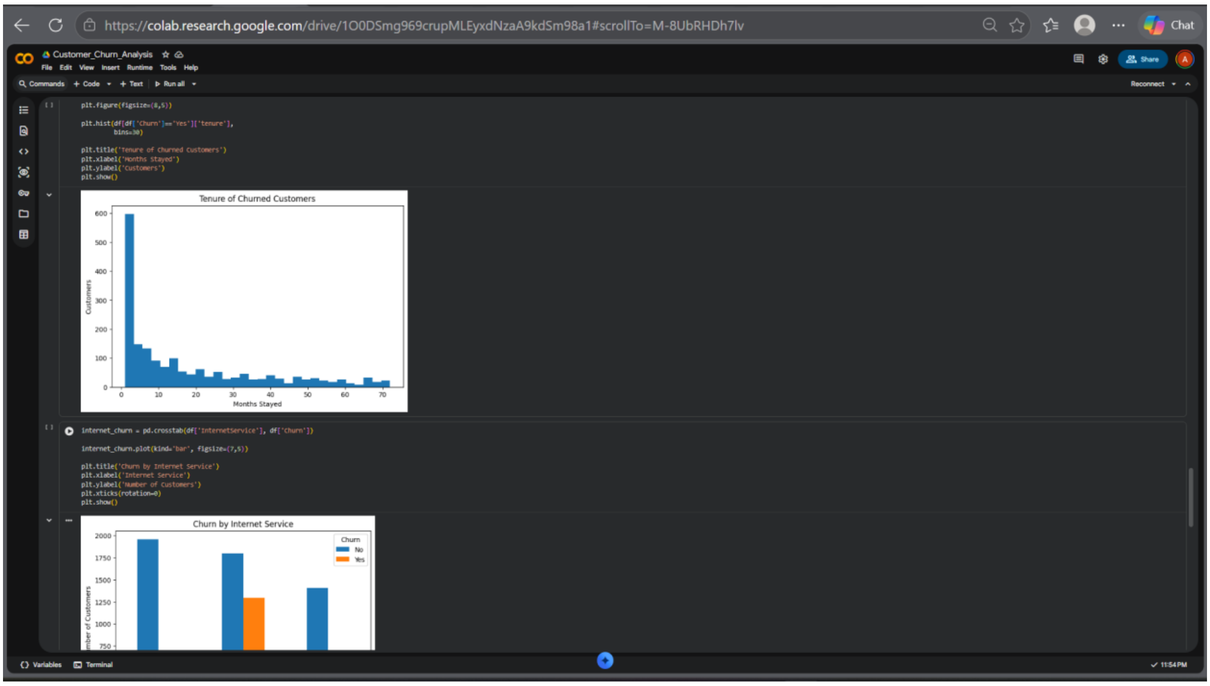
Share
Reconnect
payment_churn = pd.crosstab(df['PaymentMethod'], df['Churn'])
payment_churn.plot(kind='bar', figsize=(9,5))
plt.title('Churn by Payment Method')
plt.xlabel('Payment Method')
plt.ylabel('Number of Customers')
plt.xticks(rotation=45)
plt.show()
Churn by Payment Method
Churn
No
Yes
1200
1000
800
600
400
200
0
Number of Customers
Bank transfer (automatic)
Credit card (automatic)
Electronic check
Mailed check
Payment Method
plt.figure(figsize=(8,5))
plt.hist(df[df['Churn']=='Yes']['tenure'],
bins=30)
plt.title('Tenure of Churned Customers')
plt.xlabel('Months Stayed')
plt.ylabel('Customers')
plt.show()
Tenure of Churned Customers
600
500
Variables
Terminal
11:54 PM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Chat
Customer_Churn_Analysis
File Edit View Insert Runtime Tools Help
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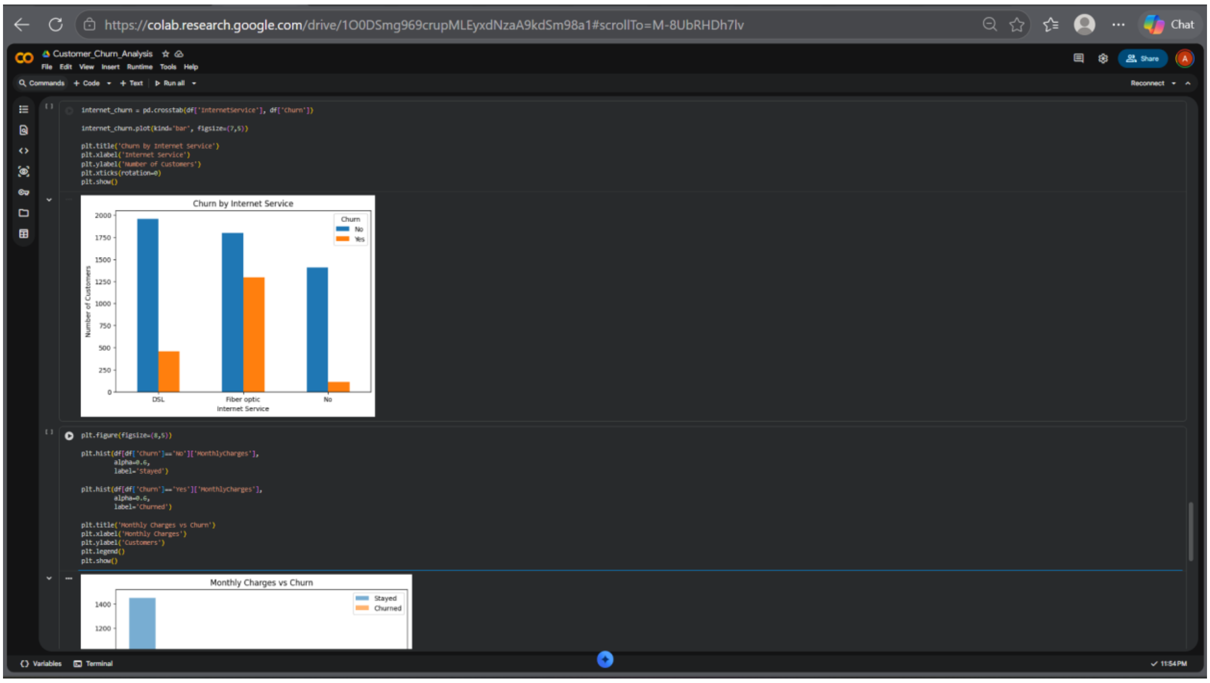
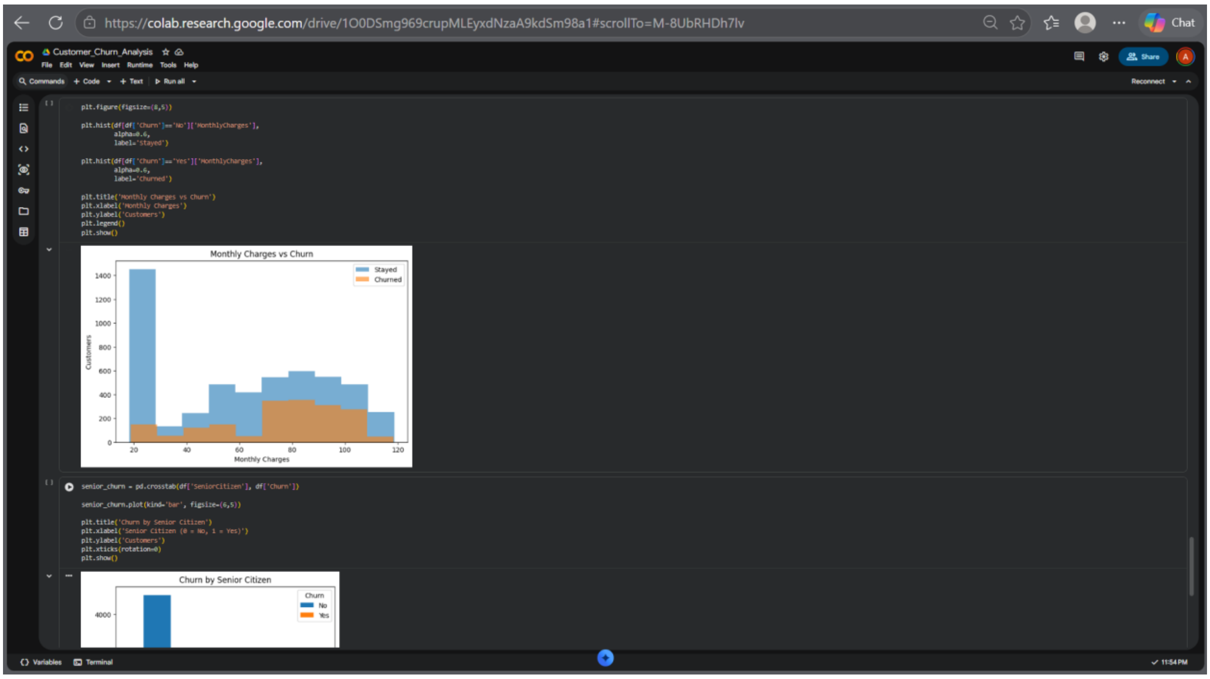
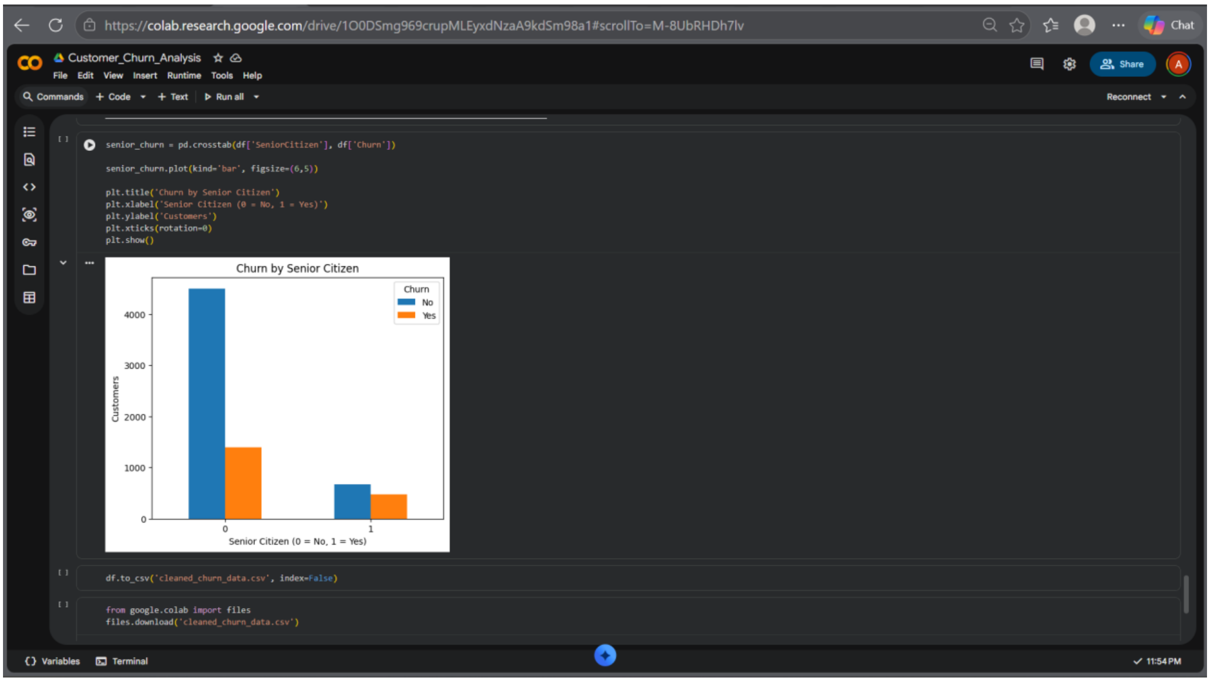
Share
Reconnect
plt.figure(figsize=(8,5))
plt.hist(df[df['Churn']=='Yes']['tenure'],
bins=30)
plt.title('Tenure of Churned Customers')
plt.xlabel('Months Stayed')
plt.ylabel('Customers')
plt.show()
Tenure of Churned Customers
600
500
400
300
200
100
0
Customers
0
10
20
30
40
50
60
70
Months Stayed
internet_churn = pd.crosstab(df['InternetService'], df['Churn'])
internet_churn.plot(kind='bar', figsize=(7,5))
plt.title('Churn by Internet Service')
plt.xlabel('Internet Service')
plt.ylabel('Number of Customers')
plt.xticks(rotation=0)
plt.show()
Churn by Internet Service
Churn
No
Yes
2000
1750
1500
1250
1000
750
Number of Customers
Variables
Terminal
11:54 PM


https://colab.research.google.com/drive/1O0DSmg969crupMLEyxdNzaA9kdSm98a1#scrollTo=M-8UbRHDh7lv
Customer_Churn_Analysis
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
internet_churn = pd.crosstab(df['InternetService'], df['Churn'])
internet_churn.plot(kind='bar', figsize=(7,5))
plt.title('Churn by Internet Service')
plt.xlabel('Internet Service')
plt.ylabel('Number of Customers')
plt.xticks(rotation=0)
plt.show()
Churn by Internet Service
Churn
No
Yes
Number of Customers
DSL
Fiber optic
No
Internet Service
plt.figure(figsize=(8,5))
plt.hist(df[df['Churn']=='No']['MonthlyCharges'],
alpha=0.6,
label='Stayed')
plt.hist(df[df['Churn']=='Yes']['MonthlyCharges'],
alpha=0.6,
label='Churned')
plt.title('Monthly Charges vs Churn')
plt.xlabel('Monthly Charges')
plt.ylabel('Customers')
plt.legend()
plt.show()
Monthly Charges vs Churn
Stayed
Churned
Variables
Terminal


https://colab.research.google.com/drive/1O0DSmg969crupMLEyxdNzaA9kdSm98a1#scrollTo=M-8UbRHDh7lv
Customer_Churn_Analysis
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
plt.figure(figsize=(8,5))
plt.hist(df[df['Churn']=='No']['MonthlyCharges'],
alpha=0.6,
label='Stayed')
plt.hist(df[df['Churn']=='Yes']['MonthlyCharges'],
alpha=0.6,
label='Churned')
plt.title('Monthly Charges vs Churn')
plt.xlabel('Monthly Charges')
plt.ylabel('Customers')
plt.legend()
plt.show()
Monthly Charges vs Churn
Stayed
Churned
Customers
Monthly Charges
senior_churn = pd.crosstab(df['SeniorCitizen'], df['Churn'])
senior_churn.plot(kind='bar', figsize=(6,5))
plt.title('Churn by Senior Citizen')
plt.xlabel('Senior Citizen (0 = No, 1 = Yes)')
plt.ylabel('Customers')
plt.xticks(rotation=0)
plt.show()
Churn by Senior Citizen
Churn
No
Yes
Variables
Terminal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Customer_Churn_Analysis
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Reconnect
senior_churn = pd.crosstab(df['SeniorCitizen'], df['Churn'])
senior_churn.plot(kind='bar', figsize=(6,5))
plt.title('Churn by Senior Citizen')
plt.xlabel('Senior Citizen (0 = No, 1 = Yes)')
plt.ylabel('Customers')
plt.xticks(rotation=0)
plt.show()
Churn by Senior Citizen
Churn
No
Yes
Customers
Senior Citizen (0 = No, 1 = Yes)
df.to_csv('cleaned_churn_data.csv', index=False)
from google.colab import files
files.download('cleaned_churn_data.csv')
Variables
Terminal
11:54 PM

Exploratory Data Analysis

Several visual analyses were performed to understand customer churn behavior.

1. Churn by Contract Type

The analysis showed that customers with **Month-to-Month contracts** had the highest churn rates compared to customers with yearly contracts.

Customers with **One-Year and Two-Year contracts** showed significantly lower churn, indicating better retention.

2. Churn by Internet Service

Customers using **Fiber Optic internet service** showed higher churn rates than DSL users.

This suggests possible dissatisfaction related to pricing, performance, or customer expectations.

3. Churn by Payment Method

Customers using **Electronic Check payment method** showed the highest churn.

This indicates that payment preferences may influence customer retention.

4. Monthly Charges vs Churn

Customers with **higher monthly charges** were observed to have greater churn probability.

This suggests price sensitivity among customers.

5. Senior Citizen Analysis

Senior citizens showed relatively higher churn behavior compared to non-senior customers.

6. Customer Tenure Analysis

Most customer churn occurred during the **first few months of subscription**.

Long-term customers were more likely to stay.

Dashboard Overview

An interactive Power BI dashboard was developed to visualize customer churn behavior effectively.

The dashboard includes:

KPI Cards

- Total Customers
- Churned Customers
- Churn Rate (%)
- Average Customer Tenure

Visual Analysis

- Churn by Contract Type
- Churn by Internet Service
- Churn by Payment Method
- Monthly Charges vs Churn
- Tenure of Churned Customers
- Churn by Senior Citizen

Interactive Features

The dashboard includes slicers for:

- Contract Type

- Internet Service
- Payment Method

These slicers allow dynamic filtering and interactive exploration of churn behavior.

The dashboard was designed using a dark-themed professional layout for improved readability and business presentation.

Key Insights

Based on the analysis, the following insights were identified:

1. Month-to-Month Contracts Have Highest Churn

Customers with monthly contracts showed the highest churn risk.

2. Fiber Optic Users Churn More

Fiber optic internet users demonstrated significantly higher churn compared to DSL users.

3. Electronic Check Customers Are High-Risk

Customers using electronic check payment methods were more likely to discontinue services.

4. Early Subscription Period Is Critical

Most customers churned during the initial months of subscription.

5. Higher Charges Increase Churn

Customers paying higher monthly charges showed greater probability of churn.

6. Senior Citizens Require Better Retention

Senior citizen customers displayed relatively higher churn behavior.

Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Promote Long-Term Contracts

Offer discounts and loyalty benefits for yearly contracts to reduce churn.

2. Improve Fiber Optic Service Experience

Investigate pricing, service quality, and customer complaints for fiber optic users.

3. Payment Method Incentives

Encourage auto-payment methods by providing cashback or discounts.

4. Improve Customer Onboarding

Provide better onboarding and support during the first few months of subscription.

5. Pricing Optimization

Introduce flexible pricing plans to reduce churn among price-sensitive customers.

6. Personalized Retention Campaigns

Design targeted offers for high-risk customer groups such as senior citizens.

Conclusion

This project successfully analyzed customer churn behavior using the Telco Customer Churn Dataset.

Through exploratory data analysis and dashboard visualization, major churn drivers such as contract type, internet service, payment methods, monthly charges, and tenure were identified.

The analysis revealed that month-to-month customers, fiber optic users, and electronic check customers were more likely to churn.

The interactive Power BI dashboard provided meaningful business insights and enabled stakeholder-friendly decision-making.

This project demonstrates the practical application of data analytics in solving real-world business problems and improving customer retention.

Future Scope

This project can be enhanced further by:

- Applying machine learning models for churn prediction.
- Building customer risk scoring systems.
- Performing cohort analysis for deeper retention tracking.
- Adding customer sentiment analysis.
- Deploying the dashboard online for real-time business monitoring.

The future implementation of predictive analytics can help businesses proactively reduce churn and improve customer retention strategies.